Protein Chemistry and Enzymology

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Protein Chemistry and Enzymology (A06217)

Short Title: Protein Chemistry & Enzymology
Faculty Science and Computing
Department Science
Cluster Biology
Author OODONOVAN
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module will provide students with a detailed description of protein and enzyme structure and function and also examines protein/enzyme purification and analysis.

Programmes

CHEM-0023	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	3 / 5 / M
CHEM-0023	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	3 / 5 / M

Indicative Content

- Primary, secondary, tertiary and quaternary structure of proteins
- Denaturation and precipitation of proteins
- Protein and enzyme extraction and purification methods
- Principles of enzyme catalysis: substrate specificity, chemical nature of enzyme catalysis, cofactors
- Enzyme inhibition and clinical inhibitors
- Enzyme technology: industrial applications, immobilisation and biosensors

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Communicate the various levels of protein structure.
2. Examine a variety of extraction and purification methods used for proteins/enzymes.
3. Explore the catalytic principles and the biochemistry of enzyme action.
4. Compare and contrast different types of enzyme inhibition and determine the importance of enzyme inhibitors in pharmacology and biochemistry.
5. Analyse and interpret data obtained from experimental analysis of proteins and enzymes and communicate these effectively.

Learning and Teaching Methods

- Lectures.
- Library work (self-directed study).
- Practical laboratory work.
- Student engagement will be enhanced through a blended teaching approach, combining on-line learning through interactive use of the Moodle virtual e-learning environment with lectures and in-class discussions.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4
Continuous Assessment	40%	
Lab Report	40%	5

Assessment Criteria

<40%: Little understanding of the basic protein chemistry and enzymology. Unable to carry out critical thinking or practical skills.

40%-49%: Is able to understand basic protein chemistry and enzymology. May have some difficulty with critical thinking and problem solving.

50%-59%: All of the above including a good understanding of theoretical protein chemistry and enzymology and demonstration of good practical skills.

60%-69%: Show a high level of competence and efficiency in theoretical and practical protein chemistry and enzymology.

70%-100%: All of the above to an excellent level. The ability to describe and discuss theoretical and practical protein chemistry, enzymology and show initiative.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- "Moodle used to provide links to useful websites." https://moodle.wit.ie/login/index.php
- Nelson, D.L. and M.M. Cox. _Lehninger Principles of Biochemistry_. 3rd Edition. NY, USA: Worth, 2000.

Supplementary Material

- "Lehninger Principles of Biochemistry." http://www.academia.edu/23720562/Lehninger_Principles_of_Biochemistry_6th_Edition_by_David_L_Nelson_and_Michael_M_Cox
- "Question bank of Biochemistry." http://site.ebrary.com/lib/waterfordit/detail.action?docID=10355534
- Matthews, C.K., K.E. van Holde, D.R. Appling and J.A. Spencer. _Biochemistry_. 4th ed. USA: Pearson-Prentice Hall, 2013.
- Nelson, D.L. and M.M. Cox. _Lehninger Principles of Biochemistry_. 6th Edition. NY, USA: Worth, 2013.
- Voet, D., J.G. Voet and C.W. Pratt. _Fundamentals of Biochemsitry_. 4th Edition. USA: Wiley & Sons Inc, 2012.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab